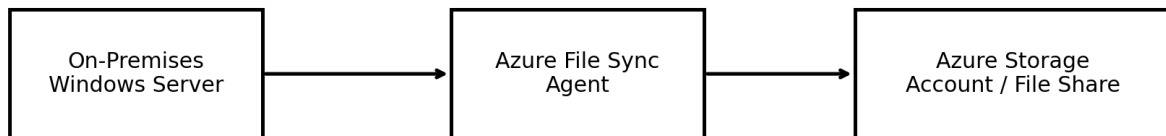


PROJECT REQUIREMENTS DOCUMENT

Azure File Sync with an On-Premises File Server

Team T175 | Project P073



File synchronization and cloud storage flow

1. Introduction

This document defines the functional, technical, infrastructure, security, and validation requirements for the project “Azure File Sync with an On-Premises File Server.” The source abstract states that the solution connects a local file server with Microsoft Azure cloud storage to improve file management, storage, synchronization, tiering, and backup.

2. Problem Definition

Companies may store large amounts of data on local servers, making storage management, synchronization, and backup increasingly difficult. The proposed solution connects the on-premises file server to Azure storage so that files can be synchronized and additional copies can be maintained in the cloud.

3. Project Objectives

- Synchronize files between the local Windows Server and an Azure file share.
- Provide additional cloud storage for organizational files.
- Support cloud tiering so frequently used files can remain locally available while seldom-accessed files are stored in the cloud.
- Support synchronization of files for multiple offices or locations.
- Maintain an additional cloud copy of local files for backup purposes.

4. Functional Requirements

ID	Requirement	Description
FR-01	Azure environment	The system shall use an Azure subscription to host the required cloud resources.
FR-02	Storage account	The system shall use an Azure Storage Account for cloud-based file storage.
FR-03	Azure file share	The system shall provide an Azure File Share as the remote cloud file location.
FR-04	Storage Sync Service	The system shall use Azure Storage Sync Service to coordinate synchronization.
FR-05	Sync agent	The Windows Server/VM shall have the Azure File Sync agent installed and configured.
FR-06	File synchronization	The solution shall synchronize selected local files/folders with the Azure file share.
FR-07	Cloud storage	The solution shall allow additional files to be stored in Azure.
FR-08	Cloud tiering	The solution shall support keeping frequently accessed data locally while moving less frequently accessed data to cloud storage.
FR-09	Multiple offices	The design shall support synchronization of file data across multiple office locations where additional server endpoints are configured.
FR-10	Backup copy	The solution shall maintain an additional copy of local files in Azure as described in the project use cases.

5. Infrastructure Requirements

Component	Status	Requirement
Azure subscription	Required	Eligible Azure Free/Student subscription; use only within available credits/limits.
Windows Server / VM	Required	Use an existing/local Windows Server or VM where possible to avoid Azure VM charges.
Azure Storage Account	Required	Small Azure Storage Account for academic demonstration.
Azure File Share	Required	Small Azure File Share; keep storage and test data limited.
Storage Sync Service	Required	Required for Azure File Sync; use only for the limited project demonstration.

Azure File Sync Agent	Required	Install on the Windows Server/VM used for the demonstration.
Internet connection	Required	Existing Internet connection.

6. High-Level Architecture

The high-level solution consists of a Windows Server/VM at the local environment, the Azure File Sync agent on that server, and an Azure Storage Account containing the Azure File Share. The server connects to Azure through the available Internet connection.

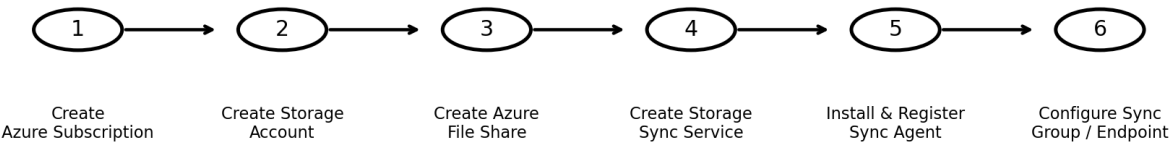


File synchronization and cloud storage flow

Figure 1. High-level architecture for the proposed Azure File Sync solution.

7. Implementation Requirements

High-level implementation sequence



- 1. Create or obtain an Azure subscription.

2. Provision an Azure Storage Account.
3. Create the required Azure File Share.
4. Create a Storage Sync Service.
5. Prepare the Windows Server/VM and install the Azure File Sync agent.
6. Register the server with the Storage Sync Service.
7. Configure the synchronization relationship and server endpoint.
8. Test file creation/modification and synchronization between the local server and Azure.
9. Validate cloud tiering and backup-related use cases described in the project.

Figure 2. Suggested high-level implementation sequence.

8. Non-Functional Requirements

Quality Attribute	Requirement
Availability	The synchronization solution should remain usable when the required Azure and local server resources are available.
Scalability	The design should allow additional file data and server endpoints to be accommodated as the organization grows.
Reliability	Synchronization should be tested under normal file creation, modification, and update scenarios.
Security	Access to Azure resources and file data should be controlled using appropriate Azure permissions and secure configuration.
Performance	File synchronization should be evaluated using representative file operations and network conditions.
Manageability	The resources and synchronization configuration should be organized so administrators can monitor and maintain them.
Recoverability	The cloud copy should support the stated backup use case and be available for recovery procedures defined during implementation.

9. Use-Case Requirements

File Sync: Users/administrators can synchronize local files with remote files in cloud storage.

Cloud Storage: Users can store additional files in Azure.

Cloud Tiering: Frequently used files remain locally accessible while seldom-accessed files can be stored in the cloud.

Multiple Offices: File data can be synchronized between different offices when corresponding endpoints are configured.

Backup: An additional cloud copy of local files can be maintained.

Free-Tier / Student Account Scope: This project is designed as a small-scale academic demonstration using an eligible Azure Free/Student subscription. Only the minimum Azure resources required for the demonstration will be used: Azure Storage Account, a small Azure File Share, Storage Sync Service, and Azure File Sync agent. Large-scale resources, additional paid services, and production-sized infrastructure are outside the project scope.

Storage capacity and test data should be kept small to minimize consumption of available Azure credits. An existing/local Windows Server or VM will be used where possible instead of creating a continuously running Azure VM.

10. Subscription Requirement and Alternatives

The project should distinguish between the Azure subscription required to create Azure resources and the payment model used for those resources. An eligible Free/Student account can provide access to an Azure subscription and available credits/limits. If a particular service or usage level is not covered by the available free allowance, the project can either use a limited academic demonstration or move to a pay-as-you-go subscription only when the additional capability is required.

Recommended academic approach: Use the Free/Student subscription for the demonstration, keep storage and test data small, and use a local Windows Server/VM. A paid subscription is useful only if the team needs resources or usage beyond the available free allowance, wants the environment to remain active after free credits expire, or needs a cloud-hosted Windows Server/VM.

11. Pay-As-You-Go Services and Project Usage

If the project is moved from a Free/Student demonstration to a Pay-As-You-Go Azure subscription, the following services/resources can be used. Pay-As-You-Go charges depend on the applicable usage meters and configuration. For this project, paid usage is mainly useful when larger storage, longer running resources, additional servers, or a more realistic multi-site demonstration is required.

Recommended PAYG scope: Azure Storage Account + Azure File Share + Storage Sync Service + one or more registered Windows Server endpoints. Add an Azure Windows VM only if a cloud-hosted server is specifically required. Azure Backup and Azure Monitor/Log Analytics are optional extensions.

Cost-control note: Azure Files PAYG usage can include storage and transactions, while Azure File Sync can generate Azure Files activity and server-registration charges. Keep test data small, avoid unnecessary continuous VM operation, and review Azure Cost Management and the Azure Pricing Calculator before extended testing.

12. Team Members

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13. Source Basis

The project title, problem statement, abstract, use cases, and original listed requirements were taken from the supplied project document. The supplied document lists Azure subscription, Windows Server/VM, Azure Storage Account, Azure File Share, Storage Sync Service, Azure File Sync agent, and Internet connection as requirements.

For this academic implementation, the scope is intentionally limited to the minimum Azure services and small test data needed for demonstration under an eligible Free/Student account. Paid or production-scale Azure infrastructure is not included.

Service / Resource	If Free Tier / Student Limits Are Enough	Alternative if Paid Usage Is Required	Benefit of Using a Paid Subscription
Azure Subscription	Use an eligible Free/Student subscription with available credit/limits.	Use Pay-As-You-Go or another institutional subscription.	Provides continued access after free credits/limits and supports larger or longer-running deployments.
Azure Storage Account	Use a small storage account and limited test data.	Use a paid storage configuration when storage or transaction requirements exceed available allowance.	Allows larger storage capacity and continued usage based on the selected pricing model.
Azure File Share	Use a small file share for demonstration and testing.	Use paid Azure Files capacity/transactions when the project requires more storage or usage.	Supports larger cloud file storage and continued operation beyond available free credits.
Storage Sync Service	Use it only for the required academic synchronization demonstration.	Use the service under a paid subscription if the project is expanded beyond the available allowance.	Enables continued Azure File Sync deployment for larger or longer-running environments.
Azure File Sync Agent	Install the agent on the local Windows Server/VM for the demonstration.	Continue using it with the required Azure resources under a paid subscription if the environment is expanded.	Supports continued synchronization in a larger or production-style deployment.
Windows Server / VM	Prefer an existing/local Windows Server or VM.	Use an Azure VM under a paid subscription if a cloud-hosted server is required.	Provides a cloud-hosted server that can be managed and scaled in Azure, but incurs VM charges.
Internet Connection	Use the existing Internet connection.	Use an appropriate organizational network connection if required.	Provides the connectivity needed for a more reliable institutional deployment; this is not an Azure free-tier service.
Azure Service / Resource	Usage in This Project	PAYG Usage / Meter	Benefit to Project
Azure Storage Account / Azure Files	Hosts the cloud file share synchronized with the Windows Server.	Used storage capacity and applicable file-share transactions.	Larger cloud repository and continued testing beyond free credits.
Azure File Share	Acts as the cloud endpoint for Azure File Sync.	Storage consumed and applicable read/write/list/other transactions.	Supports larger datasets and realistic synchronization.
Storage Sync Service	Manages registered Windows servers and sync groups.	Azure File Sync server-registration pricing and Azure Files usage generated by synchronization.	Supports continuous deployment and additional server/site endpoints.
Azure File Sync	Synchronizes local files with Azure and can provide cloud tiering.	Azure Files storage/transaction usage; tiering can create additional operations and	Demonstrates hybrid file services, multi-site sync and reduced local storage.

		data retrieval.	
Azure Windows VM (Optional)	Can host the Windows Server endpoint in Azure.	VM compute time, disks and related resources.	Provides a cloud-hosted Windows Server; optional for this project.
Azure Backup (Optional)	Can provide centralized backup/recovery for the Azure File Share.	Backup storage/snapshot and applicable backup usage.	Adds a cloud backup and restore demonstration.
Azure Monitor / Log Analytics (Optional)	Can monitor Azure resource health and activity.	Log ingestion, retention and related monitoring usage.	Adds administration, monitoring and troubleshooting capability.

14. Final Service Selection for the Academic Project

Service	Core Project?	Recommended Account
Azure Storage Account	Yes	Free/Student first; PAYG if limits are exceeded
Azure File Share	Yes	Free/Student first; PAYG for larger/longer use
Storage Sync Service	Yes	Free/Student for the demonstration; PAYG for extended use
Azure File Sync Agent	Yes	Use with a local Windows Server/VM
Windows Server / Azure VM	Windows Server required; Azure VM optional	Prefer local VM; Azure VM only if specifically required
Azure Backup	Optional	PAYG if backup extension is required
Azure Monitor / Log Analytics	Optional	PAYG if monitoring extension is required